AGENTIC-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location	Meta, OR -- station KDLS, America/Los_Angeles
Committed pair	OSM 1152747559 -> 1152748037 Meta / Meta
Facade gap	84.0 m between the two halls
Weather record	42,451 real hourly records from KDLS
Plume physics	576 steady-state solves on this site's own footprints; worst intake rise 0.2583 C at 325 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2021-01-17 (clear_cool)
Plant limit	18.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	none
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 13 of 24 hours with 2 mode changes. That is 1 more than the reactive incumbent, at 0 unsafe hours against its 0. 10 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	13 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
Incumbent	12 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
	10 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	FREE	yes	-	5.562	18.0	4.440	1.268
01	FREE	yes	-	6.154	18.0	5.127	1.401
02	FREE	yes	-	6.154	18.0	5.127	1.344
03	FREE	yes	-	5.704	18.0	4.679	1.212
04	FREE	yes	-	7.037	18.0	6.334	1.398
05	FREE	yes	-	6.950	18.0	6.110	1.210
06	FREE	yes	-	7.037	18.0	6.122	1.255
07	FREE	yes	-	10.407	18.0	9.508	1.433

08	FREE	yes	-	10.860	18.0	7.780	1.840
09	FREE	yes	-	13.442	18.0	11.178	2.116
10	FREE	yes	-	16.405	18.0	12.830	2.086
11	FREE	yes	-	16.084	18.0	13.940	2.070
12	FREE	yes	-	17.840	18.0	13.898	2.200
13	mechanical	no	dry-bulb	18.217	18.0	14.096	2.246
14	mechanical	yes	-	17.327	18.0	12.444	2.090
15	mechanical	yes	-	15.892	18.0	11.316	1.803
16	mechanical	yes	switch budget	14.873	18.0	10.149	1.759
17	mechanical	yes	switch budget	12.771	18.0	9.660	1.564
18	mechanical	yes	switch budget	11.207	18.0	9.096	1.321
19	mechanical	yes	switch budget	10.441	18.0	8.588	1.682
20	mechanical	yes	switch budget	9.283	18.0	7.378	1.733
21	mechanical	yes	switch budget	9.386	18.0	7.478	1.931
22	mechanical	yes	switch budget	9.296	18.0	7.476	1.658
23	mechanical	yes	switch budget	8.740	18.0	7.440	1.488

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 5.562 C, which is 12.438 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.268 C, and the margin is measured, not chosen: 0.986 C of group-conditional forecast error for this hour of day, plus 0.1302 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

01:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 6.154 C, which is 11.846 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.401 C, and the margin is measured, not chosen: 0.934 C of group-conditional forecast error for this hour of day, plus 0.3156 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

02:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 6.154 C, which is 11.846 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.344 C, and the margin is measured, not chosen: 0.876 C of group-conditional forecast error for this hour of day, plus 0.3156 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

03:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 5.703 C, which is 12.297 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.212 C, and the margin is measured, not chosen: 0.795 C of group-conditional forecast error for this hour of day, plus 0.2646 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

04:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.037 C, which is 10.963 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.398 C, and the margin is measured, not chosen: 1.031 C of group-conditional forecast error for this hour of day, plus 0.2151 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

05:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 6.950 C, which is 11.050 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.210 C, and the margin is measured, not chosen: 0.929 C of group-conditional forecast error for this hour of day, plus 0.1289 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

06:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.037 C, which is 10.963 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.255 C, and the margin is measured, not chosen: 0.888 C of group-conditional forecast error for this hour of day, plus 0.2146 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

07:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.407 C, which is 7.593 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.433 C, and the margin is measured, not chosen: 1.032 C of group-conditional forecast error for this hour of day, plus 0.2490 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

08:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.860 C, which is 7.140 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.840 C, and the margin is measured, not chosen: 1.545 C of group-conditional forecast error for this hour of day, plus 0.1429 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

09:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 13.442 C, which is 4.558 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.116 C, and the margin is measured, not chosen: 1.734 C of group-conditional forecast error for this hour of day, plus 0.2303 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

10:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.405 C, which is 1.595 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.086 C, and the margin is measured, not chosen: 1.680 C of group-conditional forecast error for this hour of day, plus 0.2542 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

11:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.084 C, which is 1.916 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.070 C, and the margin is measured, not chosen: 1.673 C of group-conditional forecast error for this hour of day, plus 0.2448 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

12:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.840 C, which is 0.160 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.200 C, and the margin is measured, not chosen: 1.780 C of group-conditional forecast error for this hour of day, plus 0.2684 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.217 C against a 18.0 C limit, so it fails by 0.217 C. It would take a limit 0.217 C higher, or a bound 0.217 C tighter, to change this hour.

14:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.327 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

15:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.892 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

16:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 14.873 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

17:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.771 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

18:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 11.207 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

19:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 10.441 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

20:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 9.283 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

21:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 9.385 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

22:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 9.296 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

23:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 8.740 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

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